Advanced Programming Practice · 21CSC203P

Unit 1 · Session 7 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 7 : SLO – 1 | OOP Principles

Activity (Solutions)

• 1) Encapsulation + inheritance: class Person { private String name; public void setName(String n){ name=n; } public String getName(){ return name; } } class Student extends Person { int roll; } — private field with public accessors = encapsulation; Student inherits Person's members = inheritance.

• 2) Abstraction with abstract class: abstract class Shape { abstract double area(); } class Square extends Shape { double s; Square(double s){this.s=s;} double area(){ return s*s; } } — Shape cannot be instantiated; subclasses implement the abstract method.

• 3) Polymorphism via overriding: class Animal { void sound(){ System.out.println("some sound"); } } class Dog extends Animal { @Override void sound(){ System.out.println("bark"); } } Animal a = new Dog(); a.sound(); → prints 'bark' (run-time polymorphism).